IMRS Step 09 Evaluation Report

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1. Overview

This report summarizes the biological and cross-dataset performance of the Innate Immune Response Score (IMRS) using the Step 09 split-contrast evaluation outputs.

Main questions:

1. Does IMRS separate DELIVERY from CONTROL samples?
2. Does the score generalize beyond anchor datasets?
3. Are the frozen score weights biologically plausible?

The analysis below uses Step 09 evaluation tables and the frozen anchor gene weights.

```
project_root <- "D:/IMRS_Project"

eval_tbl <- read_tsv(file.path(project_root, "05_score/transfer/eval/step09_split_eval.tsv"))
summary_tbl <- read_tsv(file.path(project_root, "05_score/transfer/eval/step09_split_summary.tsv"))
sample_tbl <- read_tsv(file.path(project_root, "05_score/transfer/eval/step09_split_sample_level.tsv"))
weights_tbl <- read_tsv(file.path(project_root, "05_score/anchors/gene_weights.tsv"))
```

```
required_eval_cols <- c("pass", "dataset_type", "gse_id", "split_id", "delta_mean_imrs_z",
    "auc_imrs_z")

missing_eval <- setdiff(required_eval_cols, names(eval_tbl))

if (length(missing_eval) > 0) {
    stop("eval_tbl missing columns: ", paste(missing_eval, collapse = ", "))
}

if (!"imrs_z" %in% names(sample_tbl)) {
    stop("sample_tbl must contain imrs_z")
}

if (!"condition_simple" %in% names(sample_tbl)) {
    stop("sample_tbl must contain condition_simple")
}
```

```
pass_eval <- eval_tbl %>%
    dplyr::filter(pass)

join_keys <- intersect(c("gse_id", "split_id", "dataset_type"), names(sample_tbl))
```

```r
if (length(join_keys) >= 2) {
    pass_sample_tbl <- sample_tbl %>%
        dplyr::inner_join(pass_eval %>%
            dplyr::select(all_of(join_keys)) %>%
            dplyr::distinct(), by = join_keys)
} else {
    pass_sample_tbl <- sample_tbl
}
```

2. Basic summary statistics

We summarize overall Step 09 performance across passing contrasts.

```r
overall_stats <- pass_eval %>%
    summarise(n_contrasts = n(), mean_delta = mean(delta_mean_imrs_z, na.rm = TRUE),
        median_delta = median(delta_mean_imrs_z, na.rm = TRUE), prop_positive_delta = mean(delta_mean_im
            0, na.rm = TRUE), mean_auc = mean(auc_imrs_z, na.rm = TRUE),
        median_auc = median(auc_imrs_z, na.rm = TRUE))

overall_stats
```

```
## # A tibble: 1 x 6
##   n_contrasts mean_delta median_delta prop_positive_delta mean_auc median_auc
##         <int>      <dbl>        <dbl>               <dbl>    <dbl>      <dbl>
## 1          66       6.12         2.88               0.939    0.901          1
```

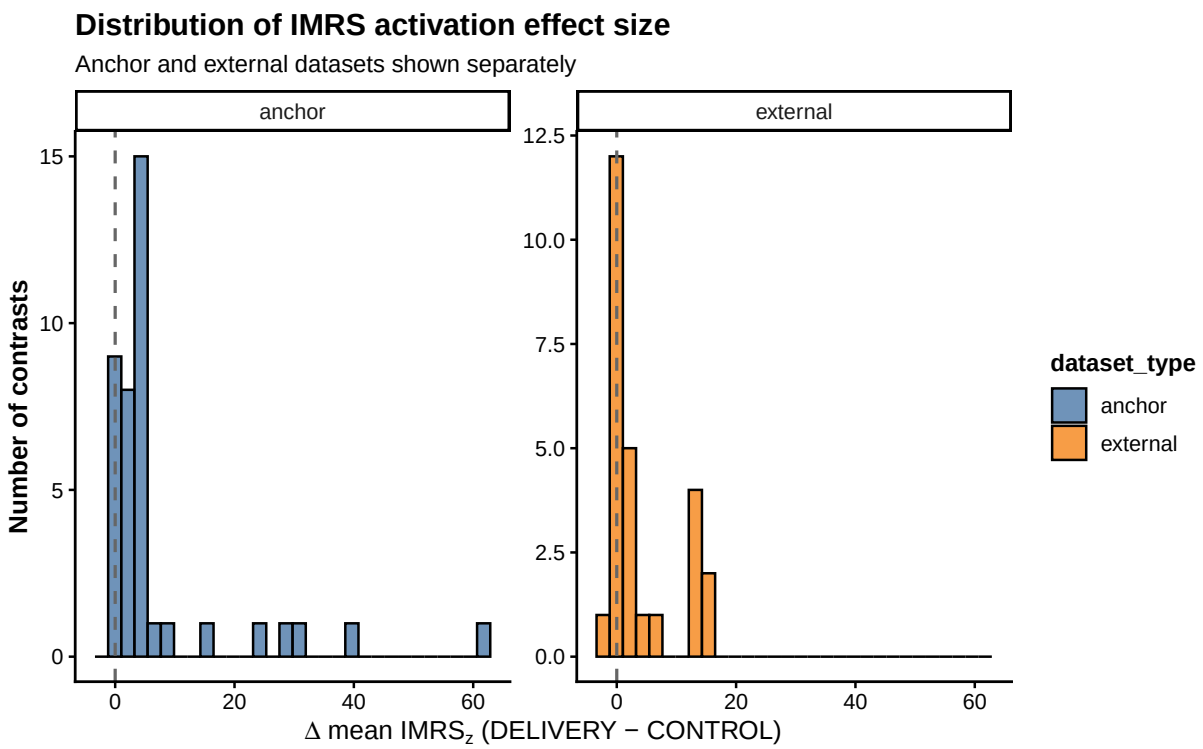
Positive `delta_mean_imrs_z` indicates DELIVERY samples have higher IMRS scores than CONTROL samples.
AUC values above 0.5 indicate separation ability.

3. Distribution of effect size across contrasts

Primary metric:

$$\Delta IMRS_z = \overline{IMRS_z(DELIVERY)} - \overline{IMRS_z(CONTROL)}$$

```r
ggplot(pass_eval, aes(x = delta_mean_imrs_z, fill = dataset_type)) + geom_histogram(bins = 30,
    alpha = 0.8, color = "black") + geom_vline(xintercept = 0, linetype = "dashed") +
    facet_wrap(~dataset_type, scales = "free_y") + scale_fill_manual(values = imrs_palette) +
    labs(title = "Distribution of IMRS activation effect size", subtitle = "Anchor and external datasets
        x = expression(Delta * " mean IMRS"[z] * " (DELIVERY - CONTROL)"),
        y = "Number of contrasts")
```

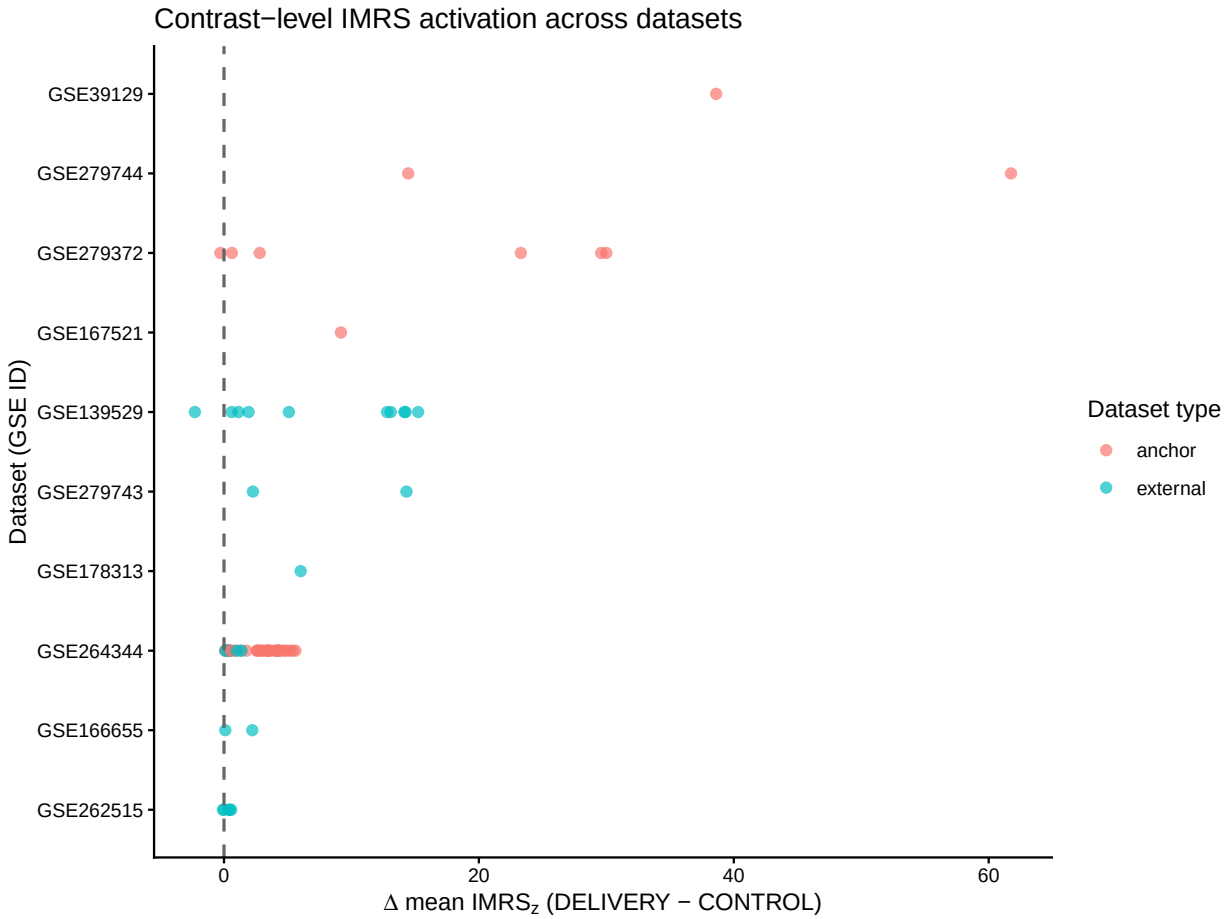


Anchor datasets contributed to construction of the IMRS score and therefore represent in-sample behavior. External datasets provide a more stringent test of score transferability. Separating the two groups avoids overstating generalization performance and makes differences between training and validation datasets easier to interpret.

3B. Contrast-level effects by dataset

This plot shows each DELIVERY–CONTROL contrast individually.
Large vertical spread within a dataset indicates heterogeneous biological responses.

```
ggplot(pass_eval, aes(x = delta_mean_imrs_z, y = reorder(gse_id, delta_mean_imrs_z,
    FUN = median), color = dataset_type)) + geom_point(alpha = 0.7, size = 2) +
    geom_vline(xintercept = 0, linetype = "dashed") + theme_classic() +
    labs(title = "Contrast-level IMRS activation across datasets", x = expression(Delta *
        " mean IMRS"[z] * " (DELIVERY - CONTROL)"), y = "Dataset (GSE ID)",
        color = "Dataset type")
```



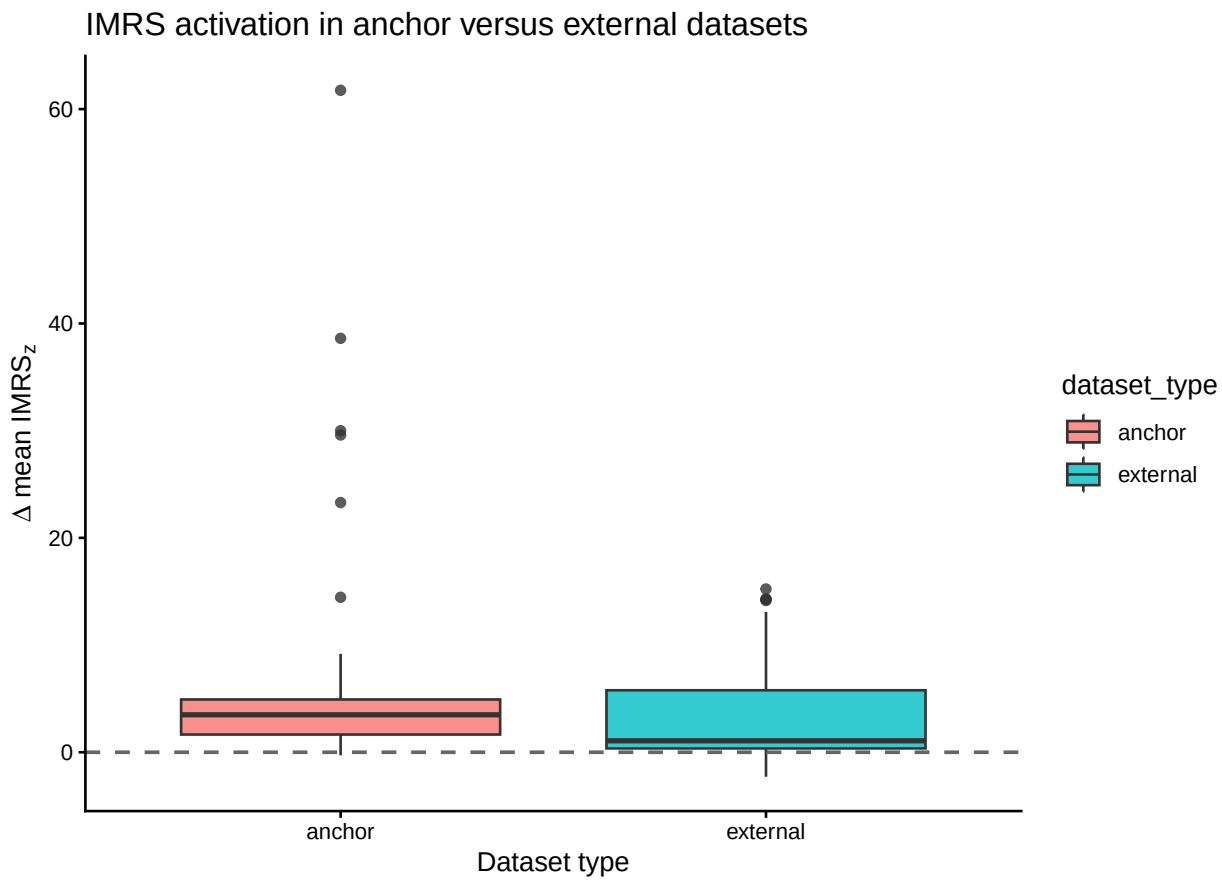
4. Anchor versus external datasets

```
anchor_external_stats <- pass_eval %>%
    group_by(dataset_type) %>%
    summarise(n_contrasts = n(), mean_delta = mean(delta_mean_imrs_z, na.rm = TRUE),
        median_delta = median(delta_mean_imrs_z, na.rm = TRUE), prop_positive_delta = mean(delta_mean_i
            0, na.rm = TRUE), mean_auc = mean(auc_imrs_z, na.rm = TRUE),
        median_auc = median(auc_imrs_z, na.rm = TRUE), .groups = "drop")

anchor_external_stats
```

```
## # A tibble: 2 x 7
##   dataset_type n_contrasts mean_delta median_delta prop_positive_delta mean_auc
##   <chr>              <int>      <dbl>        <dbl>               <dbl>    <dbl>
## 1 anchor                40       7.45         3.50               0.975    0.958
## 2 external              26       4.07         1.07               0.885    0.814
## # i 1 more variable: median_auc <dbl>
```

```
ggplot(pass_eval, aes(x = dataset_type, y = delta_mean_imrs_z, fill = dataset_type)) +
    geom_boxplot(alpha = 0.8) + geom_hline(yintercept = 0, linetype = "dashed") +
```

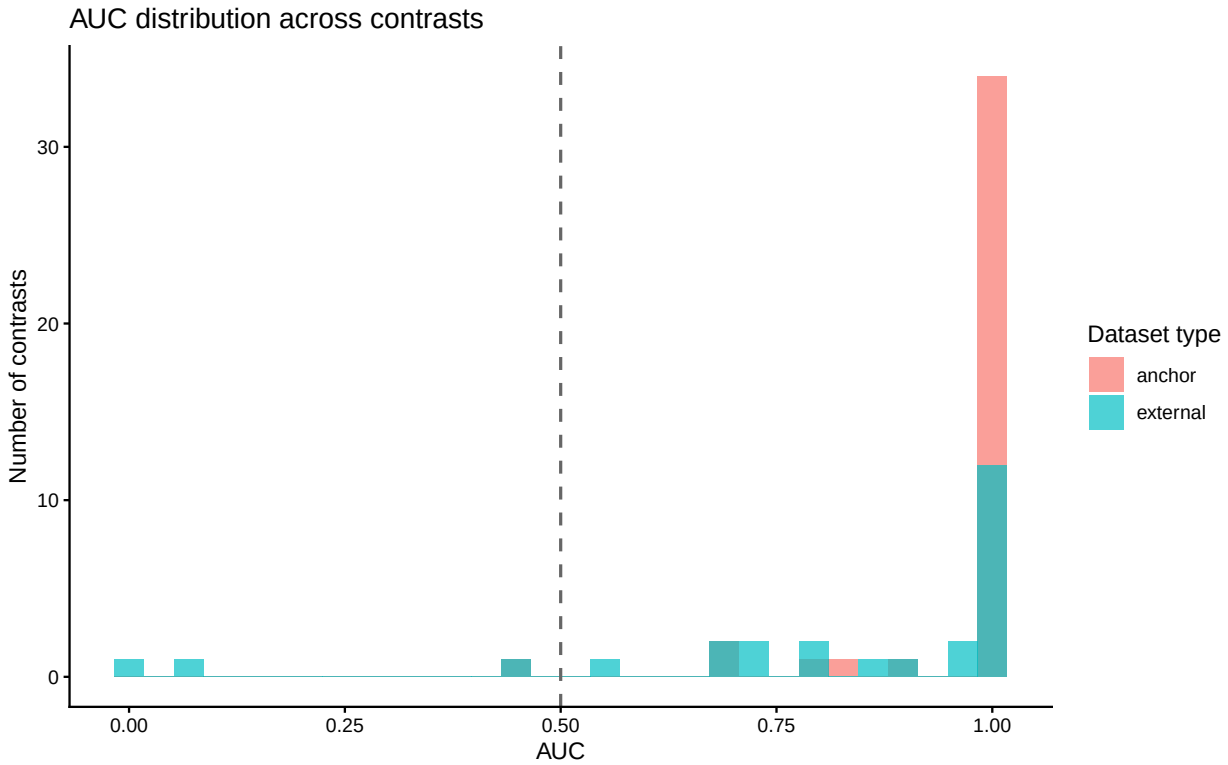
```
    theme_classic() + labs(title = "IMRS activation in anchor versus external datasets",
    x = "Dataset type", y = expression(Delta * " mean IMRS"[z]))
```



Anchors are expected to show stronger effects because they contributed to score construction.

5. Classification performance across contrasts

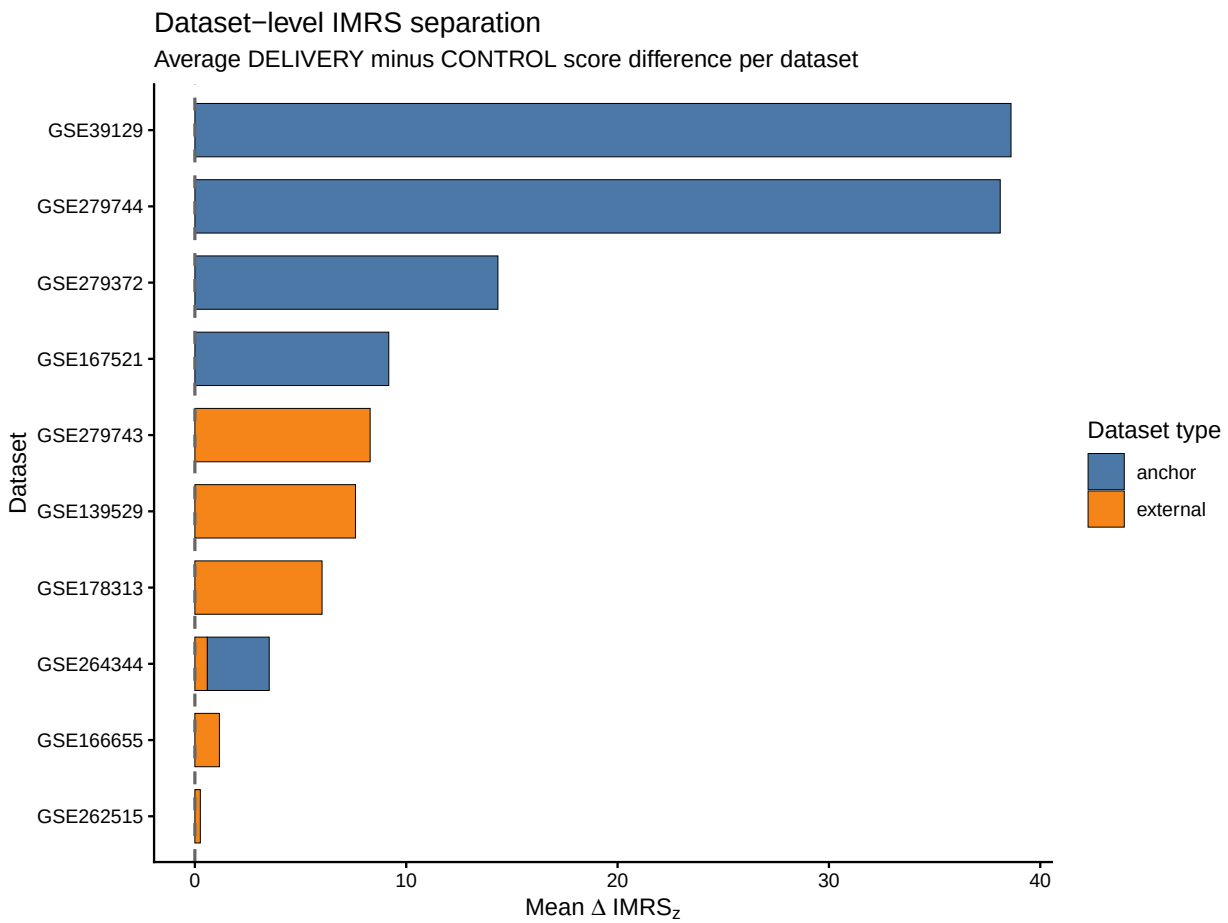
```
ggplot(pass_eval %>%
    filter(is.finite(auc_imrs_z)), aes(x = auc_imrs_z, fill = dataset_type)) +
    geom_histogram(bins = 30, alpha = 0.7, position = "identity") + geom_vline(xintercept = 0.5,
    linetype = "dashed") + theme_classic() + labs(title = "AUC distribution across contrasts",
    x = "AUC", y = "Number of contrasts", fill = "Dataset type")
```



High AUC indicates strong classification ability. However, small sample sizes may inflate AUC values.

6. dataset-level score separation

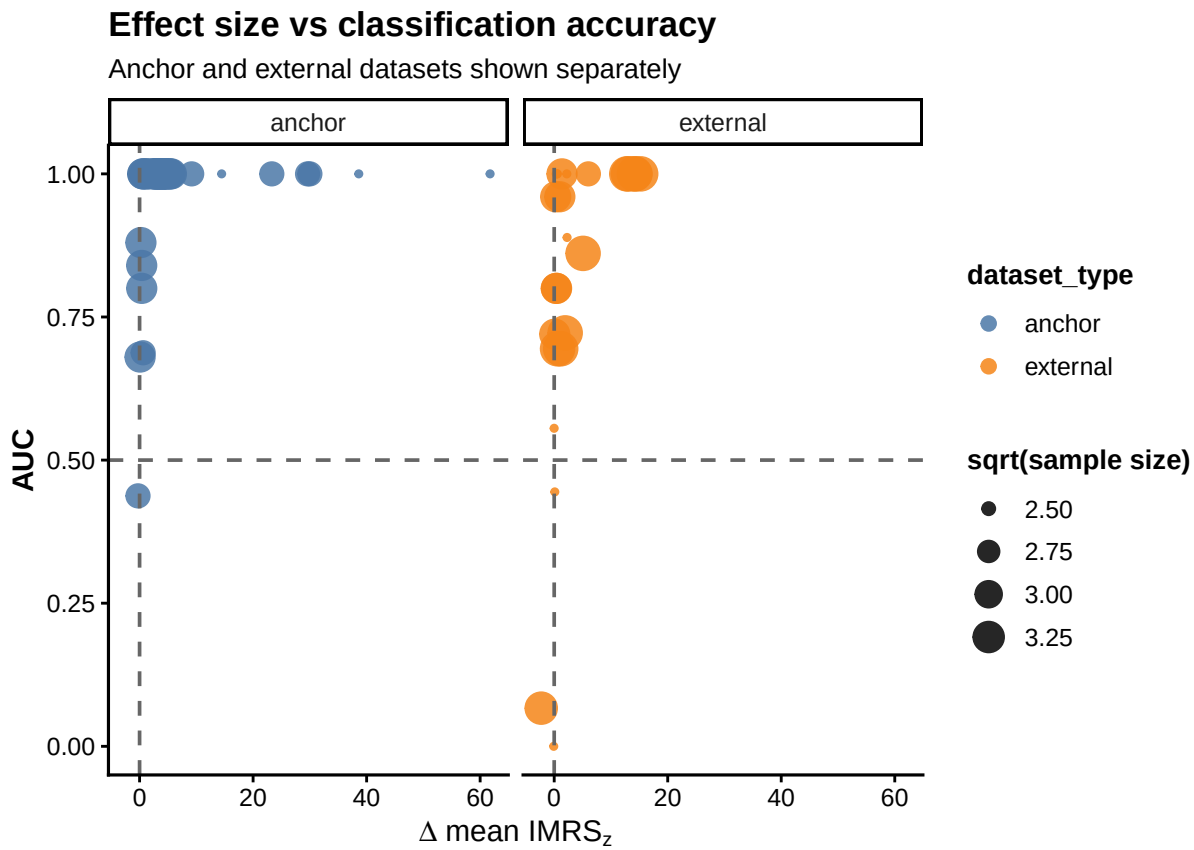
```
dataset_sep <- pass_eval %>%
    dplyr::group_by(gse_id, dataset_type) %>%
    dplyr::summarise(mean_delta = mean(delta_mean_imrs_z, na.rm = TRUE),
        .groups = "drop")

ggplot(dataset_sep, aes(x = mean_delta, y = reorder(gse_id, mean_delta),
    fill = dataset_type)) + geom_col(width = 0.7, color = "black", linewidth = 0.2) +
    geom_vline(xintercept = 0, linetype = "dashed") + scale_fill_manual(values = imrs_palette) +
    theme_classic() + labs(title = "Dataset-level IMRS separation", subtitle = "Average DELIVERY minus (
    x = expression("Mean " * Delta * " IMRS"[z]), y = "Dataset", fill = "Dataset type")
```



Because IMRS values are standardized within each dataset using dataset-specific control samples, absolute sample-level values are not directly comparable across studies. For this reason, pooled sample-level boxplots were replaced with a dataset-level summary of DELIVERY–CONTROL separation. Positive values indicate higher IMRS activation in DELIVERY samples within the same dataset.

7. Effect size versus classification strength

```
scatter_df <- pass_eval %>%
    filter(is.finite(delta_mean_imrs_z), is.finite(auc_imrs_z)) %>%
    mutate(sample_size = n_controls + n_delivery)

ggplot(scatter_df, aes(x = delta_mean_imrs_z, y = auc_imrs_z, color = dataset_type,
    size = sqrt(sample_size))) + geom_point(alpha = 0.85) + geom_vline(xintercept = 0,
    linetype = "dashed") + geom_hline(yintercept = 0.5, linetype = "dashed") +
    facet_wrap(~dataset_type) + scale_color_manual(values = imrs_palette) +
    labs(title = "Effect size vs classification accuracy", subtitle = "Anchor and external datasets show
        x = expression(Delta * " mean IMRS"[z]), y = "AUC", size = "sqrt(sample size)")
```



Both effect size (ΔIMRS$_z$) and AUC measure separation between DELIVERY and CONTROL samples. Faceting by dataset type allows direct comparison between anchor datasets used during score construction and independent external datasets used for validation. Point size reflects total sample size, highlighting contrasts supported by larger datasets.

8. Top contrasts by performance

```
# Select the single best-performing contrast from each dataset
top_contrasts <- pass_eval %>%
    group_by(gse_id) %>%
    slice_max(order_by = delta_mean_imrs_z, n = 1, with_ties = FALSE) %>%
    ungroup() %>%
    arrange(desc(delta_mean_imrs_z)) %>%
    select(dataset_type, gse_id, split_id, tissue, time_h, n_controls,
        n_delivery, delta_mean_imrs_z, auc_imrs_z) %>%
    slice_head(n = 15)

top_contrasts
```

```
## # A tibble: 10 x 9
##    dataset_type gse_id    split_id           tissue time_h n_controls n_delivery
##    <chr>        <chr>     <chr>              <chr>   <dbl>      <dbl>      <dbl>
```

```
##  1 anchor       GSE279744 GSE279744_design_~ lymph~      6          3          3
##  2 anchor       GSE39129  GSE39129_design__~ liver       4          3          3
##  3 anchor       GSE279372 GSE279372_design_~ drain~     24          4          4
##  4 external     GSE139529 GSE139529__T=bloo~ blood      24          6          6
##  5 external     GSE279743 GSE279743_design_~ Lymph~      6          3          3
##  6 anchor       GSE167521 GSE167521_design_~ skin       24          4          4
##  7 external     GSE178313 GSE178313_design_~ Liver      48          4          4
##  8 anchor       GSE264344 GSE264344_design_~ dLN         6          5          5
##  9 external     GSE166655 GSE166655_design_~ Tibia~    672          3          3
## 10 external     GSE262515 GSE262515_design_~ tissue     72          3          3
## # i 2 more variables: delta_mean_imrs_z <dbl>, auc_imrs_z <dbl>
```

9. Top weighted genes in IMRS

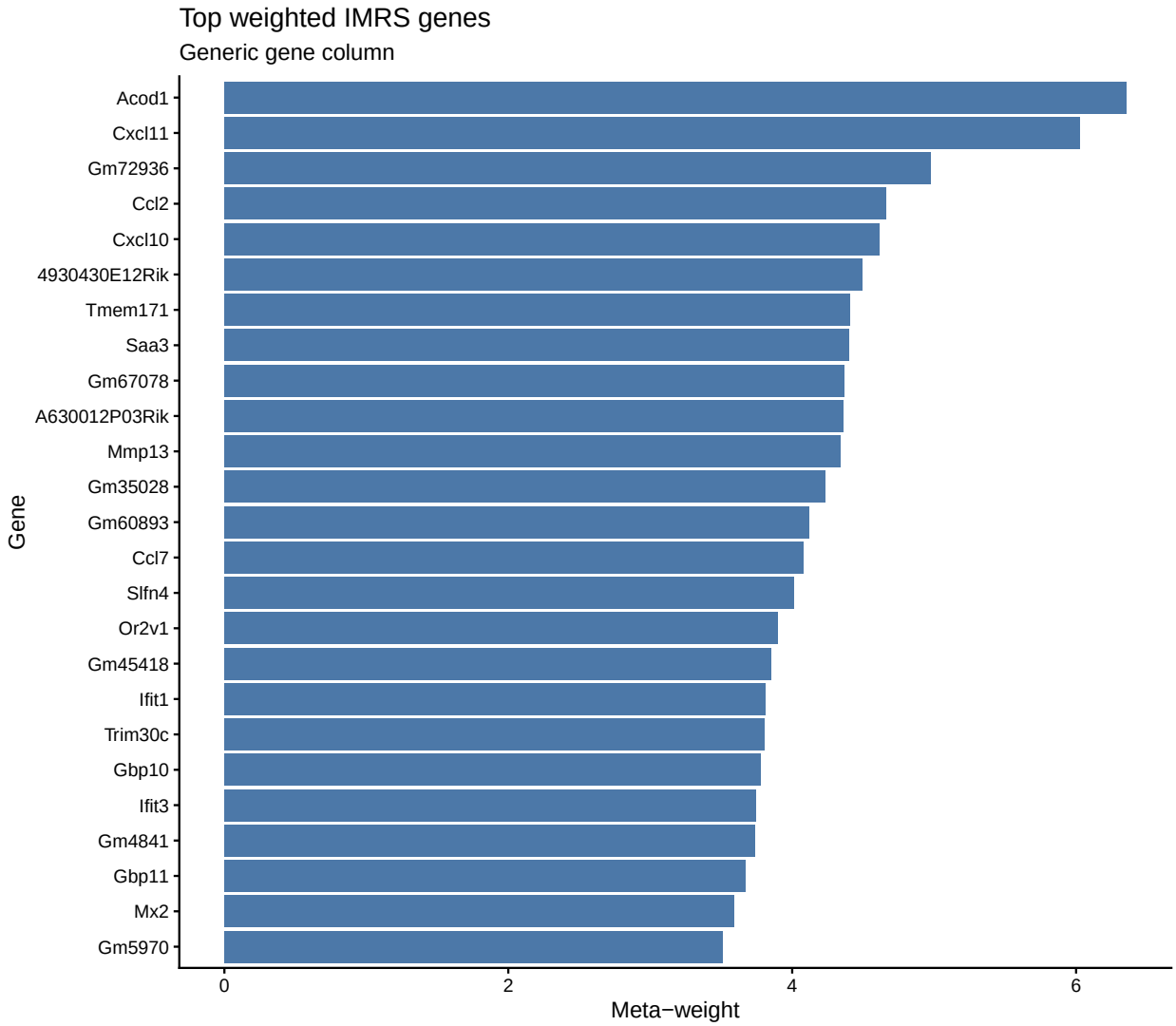
```r
# install/load biomaRt only if needed
if (!requireNamespace("biomaRt", quietly = TRUE)) {
    if (!requireNamespace("BiocManager", quietly = TRUE)) {
        install.packages("BiocManager")
    }
    BiocManager::install("biomaRt", ask = FALSE, update = FALSE)
}

library(biomaRt)

mapping_file <- file.path(project_root, "05_score/anchors/gene_symbol_mapping.tsv")

# determine gene column
if ("gene_symbol" %in% names(weights_tbl)) {
    gene_col <- "gene_symbol"
    note <- "Gene symbols"
} else if ("gene_id" %in% names(weights_tbl)) {
    gene_col <- "gene_id"
    note <- "Mapped from Ensembl IDs"
} else if ("gene" %in% names(weights_tbl)) {
    gene_col <- "gene"
    note <- "Generic gene column"
} else {
    stop("weights file missing gene column")
}

if (!"beta_meta" %in% names(weights_tbl)) {
    stop("weights file missing beta_meta")
}

top_genes <- weights_tbl %>%
    dplyr::mutate(gene_id_raw = as.character(.data[[gene_col]]), gene_id_clean = sub("\\..*$",
        "", gene_id_raw)) %>%
    dplyr::arrange(desc(abs(beta_meta))) %>%
    dplyr::slice_head(n = 25)

download_mouse_mapping <- function(gene_ids) {
```

```r
    mart <- useEnsembl(biomart = "genes", dataset = "mmusculus_gene_ensembl")

    mapping <- getBM(attributes = c("ensembl_gene_id", "mgi_symbol"), filters = "ensembl_gene_id",
        values = unique(gene_ids), mart = mart)

    mapping %>%
        dplyr::mutate(ensembl_gene_id = as.character(ensembl_gene_id),
            mgi_symbol = as.character(mgi_symbol))
}

is_ensembl <- any(grepl("^ENS", top_genes$gene_id_clean))

if (is_ensembl) {

    need_redownload <- TRUE

    if (file.exists(mapping_file)) {
        mapping_try <- tryCatch(read_tsv(mapping_file, show_col_types = FALSE),
            error = function(e) NULL)

        if (!is.null(mapping_try) && all(c("ensembl_gene_id", "mgi_symbol") %in%
            names(mapping_try))) {

            mapping_try <- mapping_try %>%
                dplyr::mutate(ensembl_gene_id = as.character(ensembl_gene_id),
                  mgi_symbol = as.character(mgi_symbol))

            if (nrow(mapping_try) > 0 && any(!is.na(mapping_try$ensembl_gene_id)) &&
                any(mapping_try$ensembl_gene_id != "")) {
                mapping <- mapping_try
                need_redownload <- FALSE
            }
        }
    }

    if (need_redownload) {
        all_gene_ids_clean <- weights_tbl[[gene_col]] %>%
            as.character() %>%
            sub("\\..*$", "", .)

        mapping <- download_mouse_mapping(all_gene_ids_clean)
        dir.create(dirname(mapping_file), recursive = TRUE, showWarnings = FALSE)
        write_tsv(mapping, mapping_file)
    }

    top_genes <- top_genes %>%
        dplyr::left_join(mapping, by = c(gene_id_clean = "ensembl_gene_id")) %>%
        dplyr::mutate(gene_label = ifelse(is.na(mgi_symbol) | mgi_symbol ==
            "", gene_id_clean, mgi_symbol))

} else {

    top_genes <- top_genes %>%
```

```r
        dplyr::mutate(gene_label = gene_id_raw)
}

top_genes %>%
    dplyr::select(gene_id_raw, gene_id_clean, gene_label, beta_meta)
```

```
## # A tibble: 25 x 4
##    gene_id_raw           gene_id_clean      gene_label    beta_meta
##    <chr>                 <chr>              <chr>             <dbl>
##  1 ENSMUSG00000022126.7  ENSMUSG00000022126 Acod1              6.35
##  2 ENSMUSG00000060183.8  ENSMUSG00000060183 Cxcl11             6.03
##  3 ENSMUSG00000138573.1  ENSMUSG00000138573 Gm72936            4.98
##  4 ENSMUSG00000035385.6  ENSMUSG00000035385 Ccl2               4.66
##  5 ENSMUSG00000034855.14 ENSMUSG00000034855 Cxcl10             4.61
##  6 ENSMUSG00000085295.9  ENSMUSG00000085295 4930430E12Rik      4.50
##  7 ENSMUSG00000052485.8  ENSMUSG00000052485 Tmem171            4.40
##  8 ENSMUSG00000040026.9  ENSMUSG00000040026 Saa3               4.40
##  9 ENSMUSG00000134421.1  ENSMUSG00000134421 Gm67078            4.37
## 10 ENSMUSG00000055110.9  ENSMUSG00000055110 A630012P03Rik      4.36
## # i 15 more rows
```

```r
ggplot(top_genes, aes(x = reorder(gene_label, beta_meta), y = beta_meta)) +
    geom_col() + coord_flip() + theme_classic() + labs(title = "Top weighted IMRS genes",
    subtitle = note, x = "Gene", y = "Meta-weight")
```



The highest-weighted IMRS genes include canonical interferon-stimulated and inflammatory response genes such as *Ifit1, Ifit3, Mx2, Cxcl10, Cxcl11, Ccl2, Ccl7, Gbp10, Gbp11,* and *Acod1*. This strongly supports the biological interpretation that IMRS captures delivery-associated innate immune activation rather than an arbitrary technical signal.

10. Mean performance by dataset

```
dataset_summary <- summary_tbl %>%
    arrange(desc(mean_delta_mean_imrs_z))

dataset_summary
```

```
## # A tibble: 16 x 7
##    dataset_type gse_id    tissue              n_contrasts mean_delta_mean_imrs_z
##    <chr>        <chr>     <chr>                     <dbl>                  <dbl>
##  1 anchor       GSE39129  liver                         1                   38.6
```

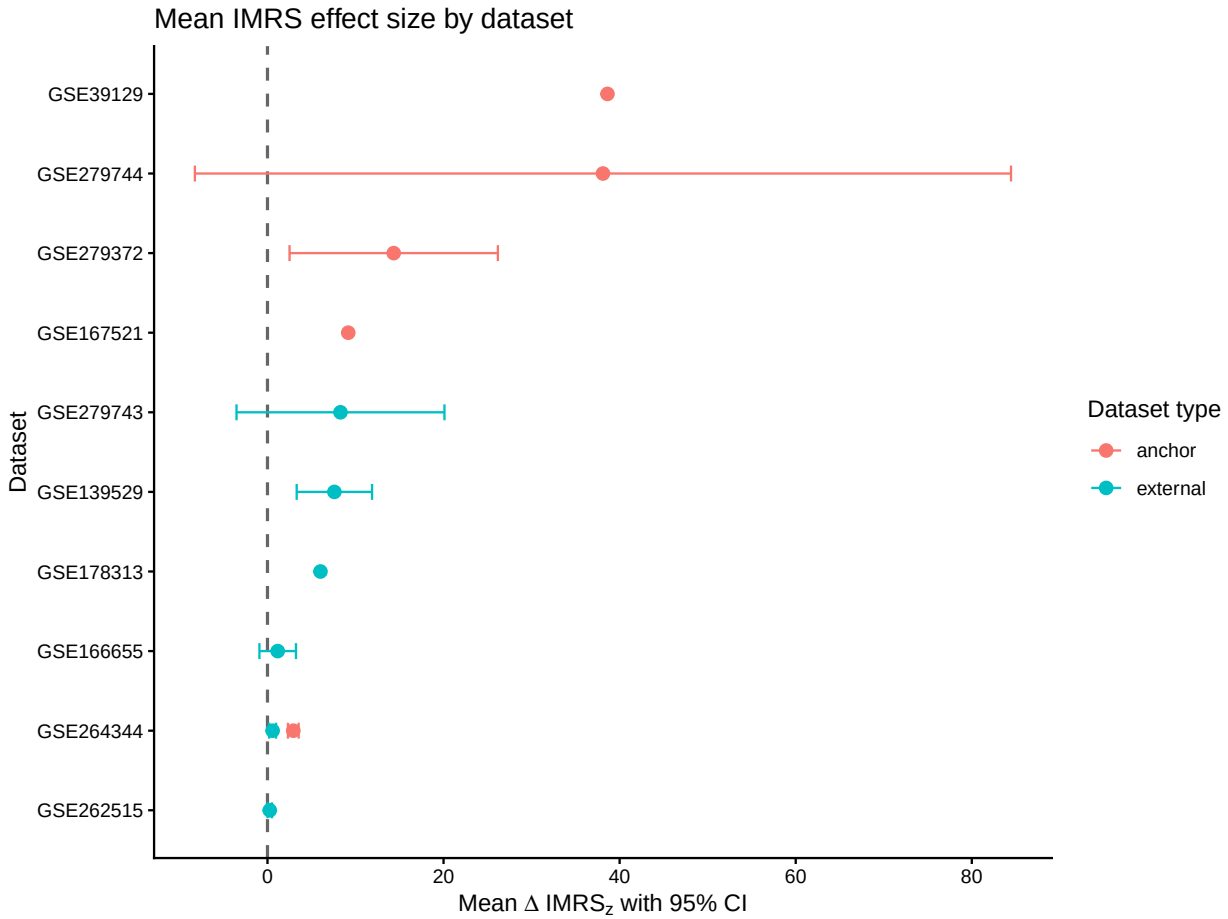
```
## 2 anchor       GSE279744 lymph node                    2                38.1
## 3 anchor       GSE279372 draining lymph nod~           6                14.3
## 4 anchor       GSE167521 skin                          1                 9.17
## 5 external     GSE279743 Lymph Node                    2                 8.29
## 6 external     GSE139529 blood                        10                 7.60
## 7 external     GSE178313 Liver                         1                 6.02
## 8 anchor       GSE264344 dLN                          10                 3.51
## 9 anchor       GSE264344 Muscle                       10                 2.90
## 10 anchor      GSE264344 Blood                        10                 2.39
## 11 external    GSE264344 Muscle                        2                 1.18
## 12 external    GSE166655 Tibialis anterior             2                 1.16
## 13 external    GSE262515 tissue                        3                 0.463
## 14 external    GSE264344 dLN                           2                 0.388
## 15 external    GSE264344 Blood                         2                 0.179
## 16 external    GSE262515 cell line                     2                -0.0477
## # i 2 more variables: median_delta_mean_imrs_z <dbl>, mean_auc_imrs_z <dbl>
```

Datasets with weaker or negative signal should be examined individually.

11. Forest-style summary by dataset

This plot summarizes average effect size by dataset with approximate 95% confidence intervals across passing contrasts.

```r
dataset_forest <- pass_eval %>%
    group_by(gse_id, dataset_type) %>%
    summarise(n_contrasts = n(), mean_delta = mean(delta_mean_imrs_z, na.rm = TRUE),
        sd_delta = sd(delta_mean_imrs_z, na.rm = TRUE), se_delta = sd_delta/sqrt(n_contrasts),
        lower = mean_delta - 1.96 * se_delta, upper = mean_delta + 1.96 *
            se_delta, .groups = "drop") %>%
    arrange(mean_delta)

dataset_forest$gse_id <- factor(dataset_forest$gse_id, levels = unique(dataset_forest$gse_id))

ggplot(dataset_forest, aes(x = mean_delta, y = gse_id, color = dataset_type)) +
    geom_vline(xintercept = 0, linetype = "dashed") + geom_errorbarh(aes(xmin = lower,
    xmax = upper), height = 0.2) + geom_point(size = 2.5) + theme_classic() +
    labs(title = "Mean IMRS effect size by dataset", x = expression("Mean " *
        Delta * " IMRS"[z] * " with 95% CI"), y = "Dataset", color = "Dataset type")
```



This plot makes it easier to see which datasets consistently support the score and which datasets are closer to zero.

12. Distribution of gene weights

This plot shows the global distribution of frozen IMRS weights.

```
ggplot(weights_tbl, aes(x = beta_meta)) + geom_histogram(bins = 40, alpha = 0.8) +
    geom_vline(xintercept = 0, linetype = "dashed") + theme_classic() +
    labs(title = "Distribution of frozen IMRS gene weights", x = "Meta-weight",
        y = "Number of genes")
```

A roughly centered distribution with tails in both directions is expected for a weighted score, while very extreme skew could indicate instability or over-dominance by a small number of genes.

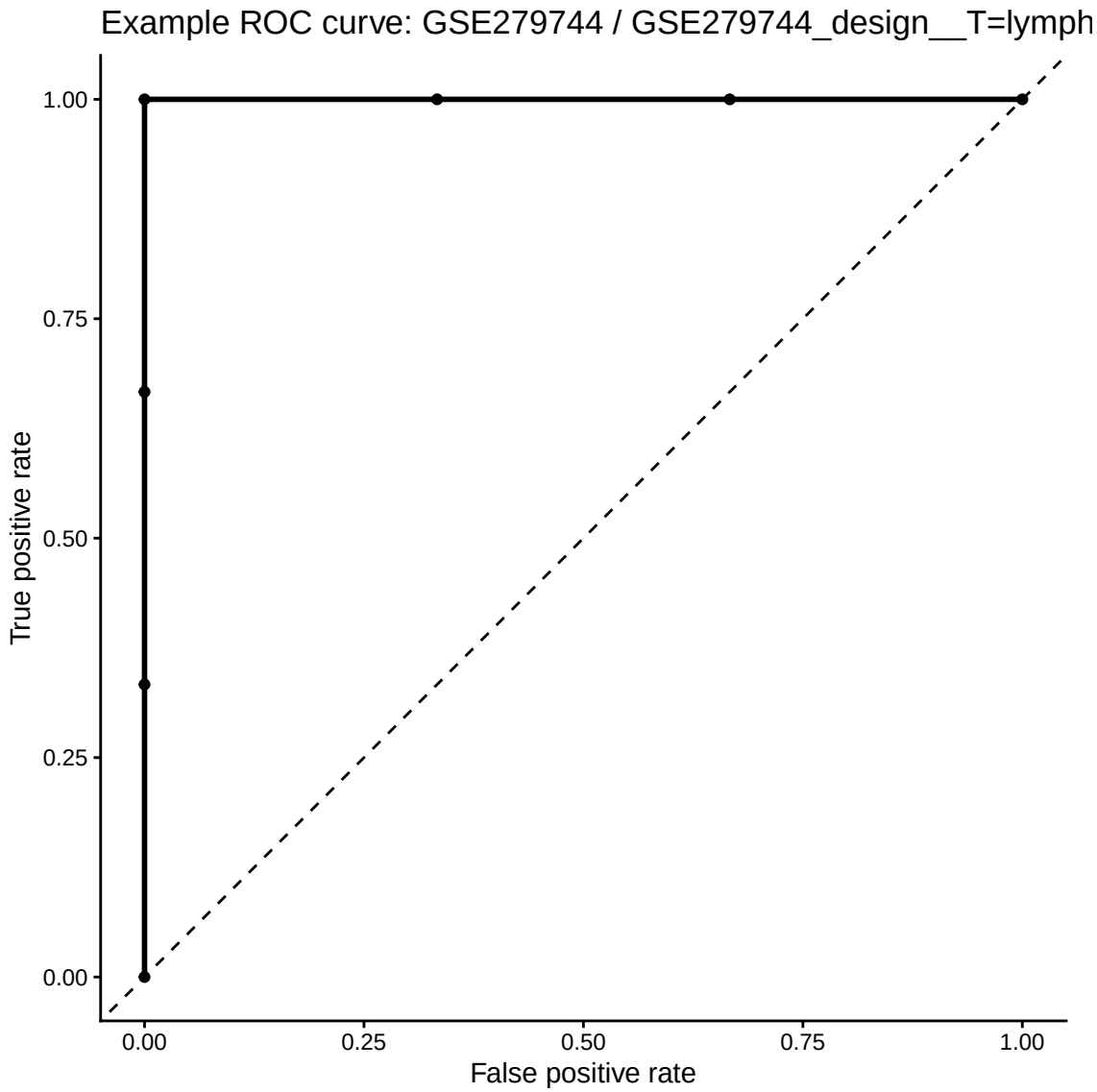
13. Example ROC curve

If sample-level truth labels are available, we can plot one example ROC curve for the best-performing passing contrast.

```
roc_ready <- all(c("gse_id", "split_id", "condition_simple", "imrs_z") %in%
    names(pass_sample_tbl))

if (roc_ready) {
    best_row <- pass_eval %>%
        filter(is.finite(auc_imrs_z)) %>%
        arrange(desc(auc_imrs_z), desc(delta_mean_imrs_z)) %>%
        slice(1)

    roc_df <- pass_sample_tbl %>%
        filter(gse_id == best_row$gse_id, split_id == best_row$split_id,
            condition_simple %in% c("CONTROL", "DELIVERY")) %>%
        mutate(truth = ifelse(condition_simple == "DELIVERY", 1, 0)) %>%
        filter(is.finite(imrs_z))
}
```

```
if (roc_ready && exists("roc_df") && nrow(roc_df) > 2 && length(unique(roc_df$truth)) ==
    2) {
```

```r
    thresholds <- sort(unique(roc_df$imrs_z), decreasing = TRUE)

    roc_curve <- lapply(thresholds, function(thr) {
        pred <- ifelse(roc_df$imrs_z >= thr, 1, 0)
        tp <- sum(pred == 1 & roc_df$truth == 1)
        fp <- sum(pred == 1 & roc_df$truth == 0)
        tn <- sum(pred == 0 & roc_df$truth == 0)
        fn <- sum(pred == 0 & roc_df$truth == 1)

        tibble(threshold = thr, tpr = ifelse((tp + fn) > 0, tp/(tp + fn),
            NA_real_), fpr = ifelse((fp + tn) > 0, fp/(fp + tn), NA_real_))
    }) %>%
        bind_rows() %>%
        distinct(fpr, tpr, .keep_all = TRUE) %>%
        arrange(fpr, tpr)

    roc_curve <- bind_rows(tibble(threshold = Inf, fpr = 0, tpr = 0), roc_curve,
        tibble(threshold = -Inf, fpr = 1, tpr = 1)) %>%
        distinct(fpr, tpr, .keep_all = TRUE) %>%
        arrange(fpr, tpr)

    ggplot(roc_curve, aes(x = fpr, y = tpr)) + geom_abline(slope = 1, intercept = 0,
        linetype = "dashed") + geom_line(linewidth = 1) + geom_point(size = 1.5) +
        coord_equal() + theme_classic() + labs(title = paste0("Example ROC curve: ",
        best_row$gse_id, " / ", best_row$split_id), x = "False positive rate",
        y = "True positive rate")

} else {
    plot.new()
    text(0.5, 0.5, "ROC example unavailable:\nrequired sample-level columns or class balance missing.")
}
```



This example provides a visual demonstration of classification behavior for one of the strongest passing contrasts.

14. Caveats

Key limitations:

1. Some contrasts have small sample sizes.
2. AUC can be unstable with small N.
3. External datasets may vary in delivery method and tissue context.
4. Gene identifiers in the weights table require mapping to symbols for biological interpretation.

15. Conclusion

The Step 09 evaluation suggests that IMRS captures delivery-associated innate immune activation across multiple studies.

Supporting evidence includes:

1. Most contrasts show positive effect size.
2. Many contrasts show AUC above 0.5.
3. DELIVERY samples tend to have higher IMRS scores than CONTROL samples.
4. External datasets retain detectable signal.

Overall, IMRS appears technically functional and shows promising transferability across datasets, although further biological validation is required. “‘